A-A-54188
15 June 1990

COMMERCIAL ITEM DESCRIPTION

REFRIGERATOR, MECHANICAL, BIOLOGICALS

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a biological refrigerator designed to contain laboratory reagents or human blood.

Salient characteristics:

The unit shall be a biological counter-top refrigerator of 5.4 cubic foot capacity designed to contain laboratory reagants or human blood. The refrigerator shall hold and maintain, at a constant temperature, 60 blood bags of 450 ml. Approximate outside dimensions of the unit shall be 36 inches high, by 24 inches wide, by 25 inches deep. The door shall be hinged on the right hand side. The refrigerator shall be capable of maintaining a preset temperature range of from 35°F to 39°F.

Standard features shall include two stainless steel drawers on nylon or other suitable rollers, dual temperature controls in case one fails, automatic defrost, and a stainless steel or equal corrosion-resistant material adapter rack for supporting blood bags. Construction shall be minimum 22 gage, 300 series stainless steel for front, sides, top, and interior of the refrigerator. The back and bottom shall be minimum 22 gage galvanized steel. The refrigerator shall have quiet, vibration-free operation.

The refrigerator shall be equipped with an audible alarm and visual temperature monitor. The monitor shall be either AC or DC operated, and shall give an instantaneous audible and visual warning when the temperature within the refrigerator falls below 33°F or rises above 42°F. The monitor shall include a fail safe switch which silences the audible alarm and signals when the safe temperature is restored. The temperature monitor shall be affixed to the refrigerator or

AMSC N/A FSC 4110

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remotely mounted and shall have a stainless steel or equal corrosion resistant material immersible temperature probe fitted to the inside of the refrigerator. The unit shall have a recording thermometer as an accessory. The recording thermometer shall be a seven day, spring-wound or electronic instrument designed to reflect minute temperature variations and shall have a replaceable eight-inch Centigrade (or Fahrenheit) temperature chart. Shall be supplied with a minimum of 50 charts.

The refrigerator shall operate on 115 volts 60 Hz, single phase current or 220 volts, 50 Hz, single phase, as specified by the customer. It shall be supplied with a UL-approved three-wire ground line cord and green dot hospital grade male plug. Where a hospital grade plug is not available, a plug suitable for use in hospital environment shall be provided.

The unit shall contain a minimum of 1/5 horsepower condensing unit. This item shall comply with all applicable requirements of the American Association of Blood Banks, the Bureau of Medical Devices and Diagnostics, and the Food and Drug Administration.

Insulation shall be foamed polyurethane or fiberglass and shall be of necessary thickness and efficiency for providing the following performance:

Ambient Temperature	Pulldown Time Hours (max)	Cycle Time (percent)
120°F	3	100
100°F	1-1/4	55
77°F	1/2	34

Internal compartment temperature shall be 39°F. Electrical power shall be 115 volts, 60 Hz for pulldown performance testing.

The self-closing door and gasket material shall be of the magnetized type or equal, which remains flexible and usable at the expected temperature of use.

Dielectric strength. The unit shall be capable of withstanding without breakdown, for one minute, a potential of 1240 volts, 60 Hz, applied between live and dead metal parts when tested as specified herein.

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Identification marking. Each refrigerator shall be permanently and legibly marked with the following:

National Stock Number.
Commercial and Government Entity.
Name or registered trademark of manufacturer.
Contract or Purchase Order Number.
Type or Model Number.
Serial Number
U. S.
Electrical Characteristics
Instructions for use (located so as to be readily visible during use).

Markings secured by adhesives shall not be used.

Wiring diagrams. A wiring diagram shall be affixed by means of an adhesive-backed mylar overlay to the back of the cabinet or to the machine compartment cover. The diagram shall portray each electrical component by sketch, outline, or symbol; and shall include the name of each component. The wiring diagram shall show the wiring to all components and any coding used for identification of wires.

Underwriters Laboratories Inc. The refrigerator shall conform to UL 471 Standard for Safety Commercial Refrigerators and Freezers. Acceptable evidence of this shall be a UL listing mark or a certified test report from an independent laboratory acceptable to the Government.

Finish. The unit shall be fabricated from materials which are inherently resistant to corrosion or which are treated to render them corrosion-resistant. Where painting or plating is used, appropriate cleaning and priming shall be performed prior to finishing. Unless otherwise specified, finish shall be equivalent or superior to that of comparable commercial equipment, and shall be free of gaps, scratches, chips, and other imperfections.

Workmanship. The refrigerator, shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One refrigerator, as specified, constitutes one unit.

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Operators and service data manuals. The contractor shall furnish manuals, handbooks and/or brochures containing complete operation, installation and service/maintenance instructions (including pictures of illustrations, as necessary) with complete schematics and wiring diagrams. All manuals, handbooks, and/or brochures will be written in the English language and all schematic and wiring diagrams will use American electrical and electronics symbols. The manuals will include electrical data and connection diagrams for all applicable utilities. The instructions shall also contain a complete list of all replaceable parts showing part number, name and quantity required. All service notes, service memos, etc., pertaining to the equipment and issued subsequent to the printing of the original instructions shall also be included. These instructions shall be, as a minimum, that furnished to service engineers (or servicemen) that normally install and service the equipment for the company or distributor. When the system being procured includes, or will operate ancillary equipment, the service data shall include complete instructions and drawings which show interfacing of all system components.

Two (2) complete sets of the above information shall be packaged with each unit of the required equipment. Where multiple quantities of the same unit are to be delivered to the same ultimate consignee, then only a total of two service manuals need to be supplied to each consignee. However, two operational manuals still need to be supplied with each unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 4 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be one refrigerator as specified in this description.

Sampling for tests. Sampling for tests including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of S-1. The unit of product for test purposes shall be one refrigerator.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Operation. The unit shall be operated for one hour according to manufacturer's directions to verify that the controls are adjusted and function. The unit shall be inspected for damage or leakage from any part, piping or fitting.

Dielectric strength. A slowly rising potential, from 0 to 1240 volts, 60 Hz, shall be applied between live and dead metal parts. The potential shall be maintained at maximum for one minute.

Pulldown performance. The unit shall obtain an internal compartment temperature of 39°F with the ambient temperatures maintained as specified herein, within the times specified, and without exceeding the percentage running times specified for at least 24 hours of steady state operation under the conditions for ambient temperature, pulldown time and cycle time specified herein.

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

4110-01-075-1239

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA - OSS
PHS
FDA-MPQAS

Project Number: 4110-0436
Location:
CID/FREEZER/H1B-22

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